

QUBIT Wind Farm Formulation

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1 Problem Statement

We have an offshore Wind farm, containing 46 Wind turbines and 2 collectors. Each collector can handle up to 23 turbines. Connecting this network with cables having different diameters.

- D_1 - 500 euro/meter can handle the power generated by 1 Turbine
- D_2 - 700 euro/meter can handle the power generated by 2 Turbines
- D_3 - 900 euro/meter can handle the power generated by 3 Turbines
- D_4 - 1100 euro/meter can handle the power generated by 4 Turbines

Each turbine can at most have 2 connections, either Turbine-Turbine + Turbine-Collector or Turbine-Turbine + Turbine-Turbine. The cables shouldn't create loops.

Objective: Minimize the cost of installing and connecting all cables.

2 Chosen Formulation

2.1 Variable

we decided to consider x_{ij}^k as our variable with:

$$x_{ij}^k = \begin{cases} 1 & : \text{Cable from } E_i \text{ to } E_j \text{ with diameter } k, E \text{ being element} \\ 0 & : \text{No cable} \end{cases}$$

The variables i , j , and k take the following values:

- $i \in [1..48]$
- $j \in [1..48]$

- $k \in [1..4]$

we also consider C_i or C_j to be a collector if $i = 47$ or $i = 48$ and j playing a symmetric role.

2.2 Cost function

As we're minimizing the cost of installing all cables in the network, it only makes sense to minimize this function with the given constraints.

$$J = \sum_k \sum_i \sum_j w_k * d_{ij} * x_{ij}^k + (\sum \lambda_i P_i(X)) \quad (1)$$

with w_k being the cost of cable with diameter k (per meter) and d_{ij} the length of said cable between Element i and j .

$P(X)$ being the penalty functions and λ the penalty coefficient.

2.3 Constraints

First constraints are the 23-turbine limit to each collector.

$$\sum_k \sum_j k * x_{c_1 t_j}^k = 23 \quad (2)$$

$$\sum_k \sum_j k * x_{c_2 t_j}^k = 23 \quad (3)$$

Then we have the tubrine connections limit.

$$\forall j, \sum_k \sum_i k * x_{ij}^k - \sum_k \sum_i k * x_{ji}^k = 1 \quad (4)$$

2.4 QUBO Vector

Since we are required to work in the QUBO formulation, which minimizes any $y^T Q y$, our variable should be transformed into a vector y .

And for each y_t we associated one x_{ij}^k such as:

$$i = \left\lfloor \frac{t}{n} \right\rfloor \text{mod } n + 1 \quad (5)$$

$$j = t \text{mod } n + 1 \quad (6)$$

$$k = \left\lfloor \frac{t}{n^2} \right\rfloor + 1 \quad (7)$$

2.5 QUBO Constraints

Each constraint will transform into a penalty function. Equation 2 and 3 will be in the form:

$$P_1(X) = \left(\sum_k \sum_j k * x_{c_1 t_j}^k - 23 \right)^2 \quad (8)$$

$$P_2(X) = \left(\sum_k \sum_j k * x_{c_2 t_j}^k - 23 \right)^2 \quad (9)$$

and Equation 4 will be:

$$P_3(X) = \sum_i \left(\sum_k \sum_j k * (x_{ij}^k - x_{ji}^k) - 1 \right)^2 \quad (10)$$

Now to get the Matrix form for this problem ($y^T Q y$) we need to develop each penalty function and get relations between components of X and then relate them to Y.

Starting with Equation 4:

$$\left(\sum_k \sum_j k * (x_{ij}^k - x_{ji}^k) - 1 \right)^2 = \left(\sum_k \sum_j k (x_{ij}^k - x_{ji}^k) \right)^2 + 1 - 2 \left(\sum_k \sum_j k (x_{ij}^k - x_{ji}^k) \right)$$

Taking the first term on its own:

$$\begin{aligned} & \left(\sum_k \sum_j k x_{ij}^k - x_{ji}^k \right)^2 = \sum_k k^2 \left[\sum_j (x_{ij}^k)^2 + 2 \sum_j \sum_{j' < j} x_{ij}^k * x_{ij'}^k \right] \\ & + 2 \sum_k \sum_{k' < k} \sum_j \sum_{j' < j} k * k' * x_{ij}^k * x_{ij'}^{k'} + \sum_k k^2 \left[\sum_j (x_{ji}^k)^2 + 2 \sum_j \sum_{j' < j} x_{ji}^k * x_{ji'}^k \right] \\ & + 2 \sum_k \sum_{k' < k} \sum_j \sum_{j' < j} k * k' * x_{ji}^k * x_{ji'}^{k'} - 2 \sum_k \sum_{k'} \sum_j \sum_{j'} k * k' * x_{ij}^k * x_{ij'}^{k'} \end{aligned} \quad (11)$$

Now the entire Penalty function corresponding to this constraint will be:

$$\begin{aligned} P_3(X) = & \sum_i \left[\sum_k k^2 \left[\sum_j (x_{ij}^k)^2 + 2 \sum_j \sum_{j' < j} x_{ij}^k * x_{ij'}^k \right] \right. \\ & + 2 \sum_k \sum_{k' < k} \sum_j \sum_{j' < j} k * k' * x_{ij}^k * x_{ij'}^{k'} \\ & + \sum_k k^2 \left[\sum_j (x_{ji}^k)^2 + 2 \sum_j \sum_{j' < j} x_{ji}^k * x_{ji'}^k \right] \\ & + 2 \sum_k \sum_{k' < k} \sum_j \sum_{j' < j} k * k' * x_{ji}^k * x_{ji'}^{k'} \\ & - 2 \sum_k \sum_{k'} \sum_j \sum_{j'} k * k' * x_{ij}^k * x_{ij'}^{k'} \\ & \left. + 1 - 2 \left(\sum_k \sum_j k (x_{ij}^k - x_{ji}^k) \right) \right] \end{aligned} \quad (12)$$

And then $P_1(X)$ and $P_2(X)$ will have the same form with $\mathbf{a}$ being either equal to 1 or 2:

$$\begin{aligned}
P_a(X) = & \sum_k k^2 \left[\sum_j (x_{cat_j}^k)^2 + 2 \sum_j \sum_{j' < j} x_{cat_j}^k * x_{cat_{j'}}^k \right] \\
& + 2 \sum_k \sum_{k' < k} \sum_j \sum_{j'} k * k' * x_{cat_j}^k * x_{cat_{j'}}^k \\
& + 23^2 - 46 \sum_k \sum_j k * x_{cat_j}^k
\end{aligned} \tag{13}$$

Finally, the cost function will be:

$$\begin{aligned}
J(X) = & \sum_k \sum_i \sum_j w_k * d_{ij} * x_{ij}^k \\
+ & \sum_i [\lambda (\sum_k k^2 \left[\sum_j (x_{ij}^k)^2 + 2 \sum_j \sum_{j' < j} x_{ij}^k * x_{ij'}^k \right] \\
+ & 2 \sum_k \sum_{k' < k} \sum_j \sum_{j'} k * k' * x_{ij}^k * x_{ij'}^k \\
+ & \sum_k k^2 \left[\sum_j (x_{ji}^k)^2 + 2 \sum_j \sum_{j' < j} x_{ji}^k * x_{j'i}^k \right] \\
+ & 2 \sum_k \sum_{k' < k} \sum_j \sum_{j'} k * k' * x_{ji}^k * x_{j'i}^k \\
- & 2 \sum_k \sum_{k' < k} \sum_j \sum_{j'} k * k' * x_{ij}^k * x_{ij'}^k \\
+ & 1 - 2(\sum_k \sum_j k(x_{ij}^k - x_{ji}^k))] \\
+ & \sum_k k^2 \left[\sum_j (x_{c_1 t_j}^k)^2 + 2 \sum_j \sum_{j' < j} x_{c_1 t_j}^k * x_{c_1 t_{j'}}^k \right] \\
+ & 2 \sum_k \sum_{k' < k} \sum_j \sum_{j'} k * k' * x_{c_1 t_j}^k * x_{c_1 t_{j'}}^k \\
+ & 23^2 - 46 \sum_k \sum_j k * x_{c_1 t_j}^k \\
+ & \sum_k k^2 \left[\sum_j (x_{c_2 t_j}^k)^2 + 2 \sum_j \sum_{j' < j} x_{c_2 t_j}^k * x_{c_2 t_{j'}}^k \right] \\
+ & 2 \sum_k \sum_{k' < k} \sum_j \sum_{j'} k * k' * x_{c_2 t_j}^k * x_{c_2 t_{j'}}^k \\
+ & 23^2 - 46 \sum_k \sum_j k * x_{c_2 t_j}^k
\end{aligned} \tag{14}$$